I give you the framework

You do the job...and it can be confusing! At first :)



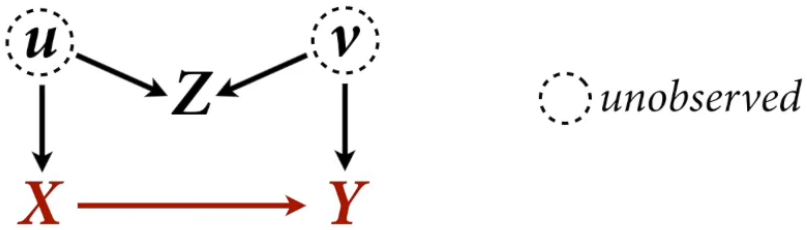
(See www.blackswanman.com)

Lecture 18: Good & Bad controls

- "Control" variable: variable introduced to an analysis so that a causal estimate is made possible.
- Common in many scientific fields because most recognise the existence of confounders. This is good.
- Common wrong heuristics for choosing control variables.
  - Anything in the spreadsheet. We will see examples why this can be a bad idea.
  - Any variables not highly collinear. Lots of different relationships in the system can create those bivariate associations. Sometimes harmless, sometimes interfere with our causal inference.
  - Any pre-treatment measurement (baseline variable). Again, we will see an example of a undesirable situation.

A first problem

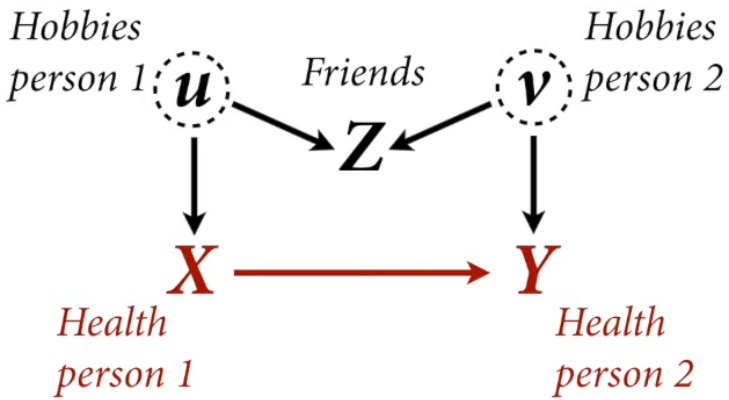
- We are going to be interested in the effect of some treatment or exposure, X on an outcome Y.
- First example:



- Z collider, while u and v forks.
- What should we do? Should we control for Z?
- Let us think of a practical example where such a situation occurs.

Most examples come from "A Crash Course in Good and Bad Controls" by Cinelli, Forney and Pearl, *Sociological Methods & Research* 2022.

A first problem: social network situation

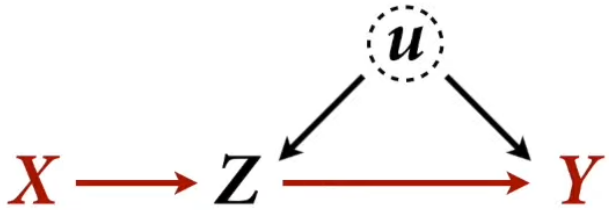


- Does the health of person 1 directly influence the health of person 2? Some sort of "social transmission".
- The unobserved variables could be any sort of lifestyle variables.
- Z is the status of "Being friends".
- Let's apply the backdoor criterion.

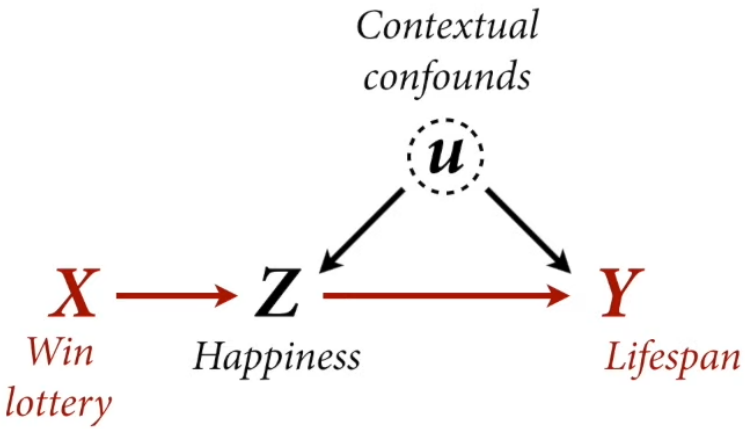
A first problem: social network situation

1. List all the paths:
   - $X \rightarrow Y$: frontdoor and open.
   - $X \rightarrow u \leftarrow Z \rightarrow v \leftarrow Y$ backdoor and closed.
2. If we add Z as a control to "be clever", it creates a problem: it opens the backdoor path.
3. Z could be a pre-treatment variable (friendship can precede the observations). It is not safe to always control pre-tretment variables, because they can share a cause with the exposure.

Example 2

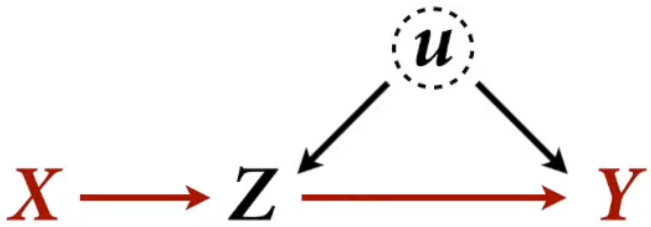


- We have a simple confound between X and Y, since Z is mediator between X and Y.
- You probably see a confound.
- If you think about a real-world example:



Example 2

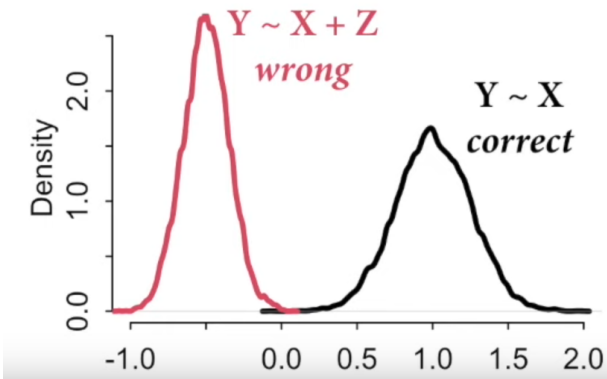
- List the paths:
  - $X \leftarrow Z \leftarrow Y$
  - $X \leftarrow Z \rightarrow u \leftarrow Y$
- No backdoor, no need to control for Z. If you do...




```
f <- function(n=100,bXZ=1,bZY=1) {
    X <- rnorm(n)
    u <- rnorm(n)
    Z <- rnorm(n, bXZ*X + u)
    Y <- rnorm(n, bZY*Z + u )
    bX <- coef( lm(Y ~ X) )['X']
    bXZ <- coef( lm(Y ~ X + Z) )['X']
    return( c(bX,bXZ) )
}

sim <- mcreplicate( 1e4 , f() , mc.cores=8 )

dens( sim[1,] , lwd=3 , xlab="posterior mean" )
dens( sim[2,] , lwd=3 , col=2 , add=TRUE )
```



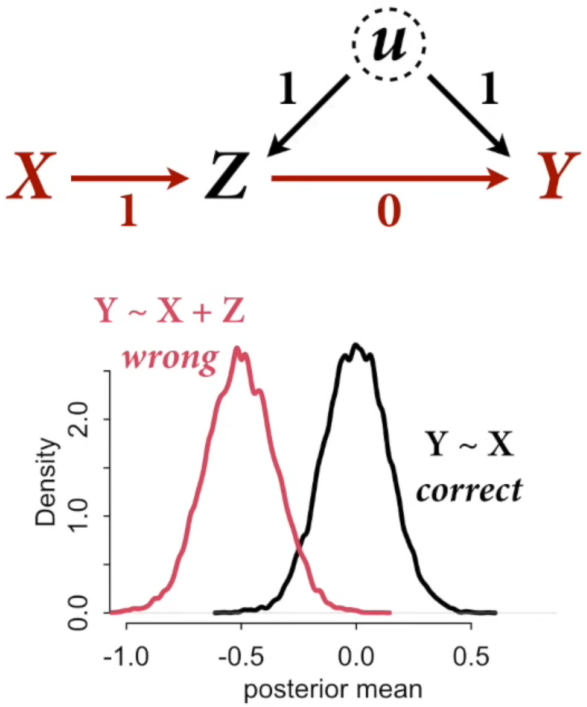
Example 2

Change bZY to zero

```
f <- function(n=100,bXZ=1,bZY=1) {
    X <- rnorm(n)
    u <- rnorm(n)
    Z <- rnorm(n, bXZ*X + u)
    Y <- rnorm(n, bZY*Z + u )
    bX <- coef( lm(Y ~ X) )['X']
    bXZ <- coef( lm(Y ~ X + Z) )['X']
    return( c(bX,bXZ) )
}

sim <- mcreplicate( 1e4 , f(bZY=0) , mc.cores=8 )

dens( sim[1,] , lwd=3 , xlab="posterior mean" )
dens( sim[2,] , lwd=3 , col=2 , add=TRUE )
```



- Message: no backdoor, no need to control for Z.
- Contolling for Z biases the treatment estimate X and opens a biasing path through u. We can estimate the effect of X, not the mediating one of Z.

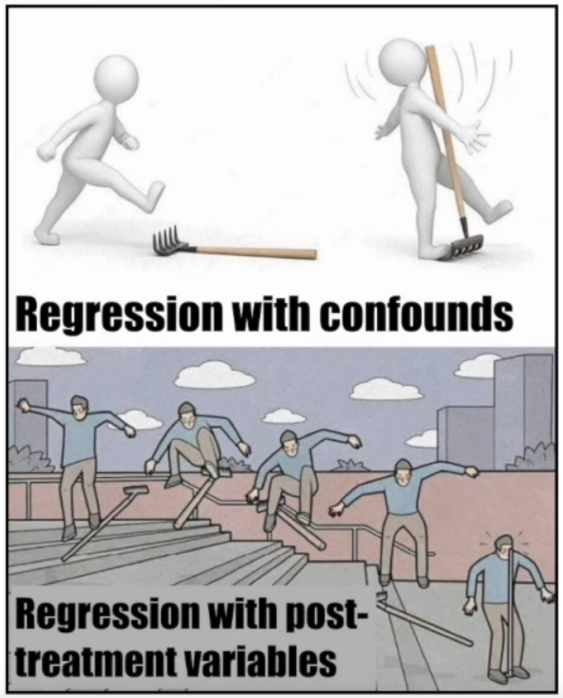
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Post-treatment bias is common

TABLE 1 Posttreatment Conditioning in Experimental Studies

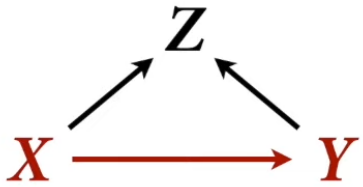
Category	Prevalence
Engages in posttreatment conditioning	46.7%
Controls for/interacts with a posttreatment variable	21.3%
Drops cases based on posttreatment criteria	14.7%
Both types of posttreatment conditioning present	10.7%
No conditioning on posttreatment variables	52.0%
Insufficient information to code	1.3%

Note: The sample consists of 2012–14 articles in the *American Political Science Review*, the *American Journal of Political Science*, and the *Journal of Politics* including a survey, field, laboratory, or lab-in-the-field experiment (n = 75).

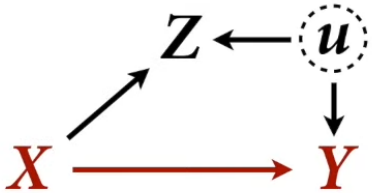


"How conditioning on Posttreatment Variables can Ruin Your Experiment", Montgomery et al. 2018

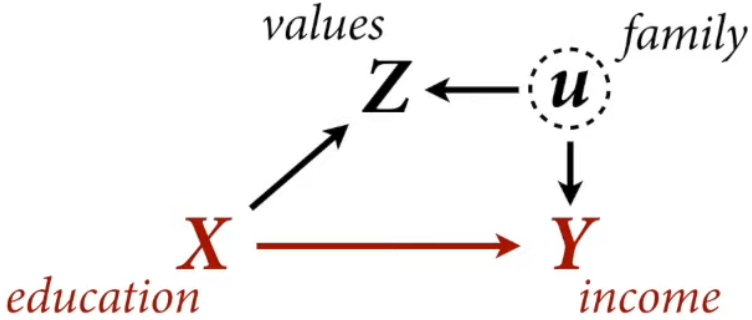
Do not touch the collider!



But colliders can be lurking:



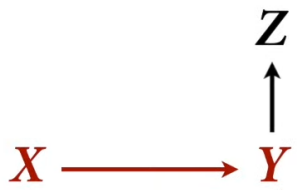
Real-life example of the previous "tricky" collider



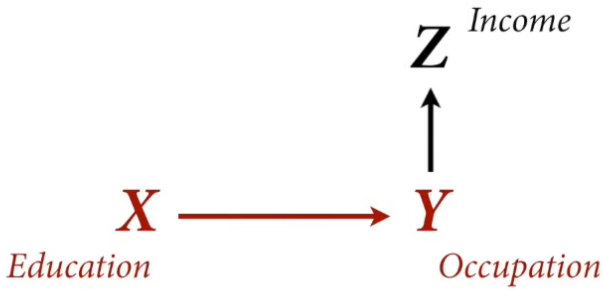
What happens if you condition on Z?

A more elaborate example

Case-control bias



- What could possibly go wrong in this pipe? Do we need to add Z? The answer is "No". Otherwise you end up removing variation in Y that Z shared with Y.
- Real life situation:



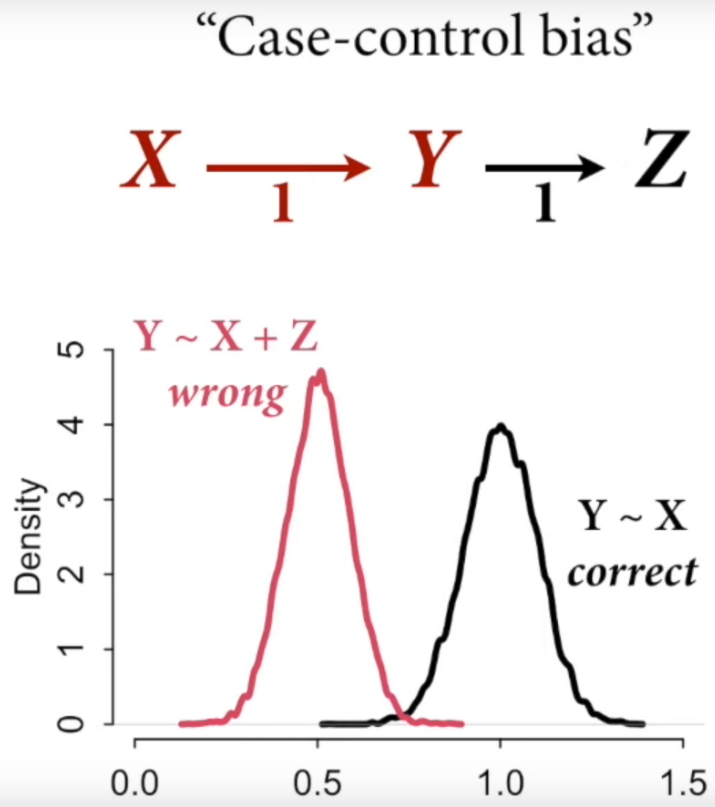
Case-control bias

A simulation

```r
f <- function(n=100,bXY=1,bYZ=1) {
    X <- rnorm(n)
    Y <- rnorm(n, bXY*X )
    Z <- rnorm(n, bYZ*Y )
    bX <- coef( lm(Y ~ X) )['X']
    bXZ <- coef( lm(Y ~ X + Z) )['X']
    return( c(bX,bXZ) )
}

sim <- mcreplicate( 1e4 , f() , mc.cores=8 )

dens( sim[1,] , lwd=3 , xlab="posterior mean" )
dens( sim[2,] , lwd=3 , col=2 , add=TRUE )
```



Z is not a cause of *Y*, but the statistical model you (wrongly) build doesn't know that.

What about the other end?

"Precise parasite"

- No backdoor paths although an arrow enters X.
- Still good not to condition on Z. This goes beyond the backdoor criterion.



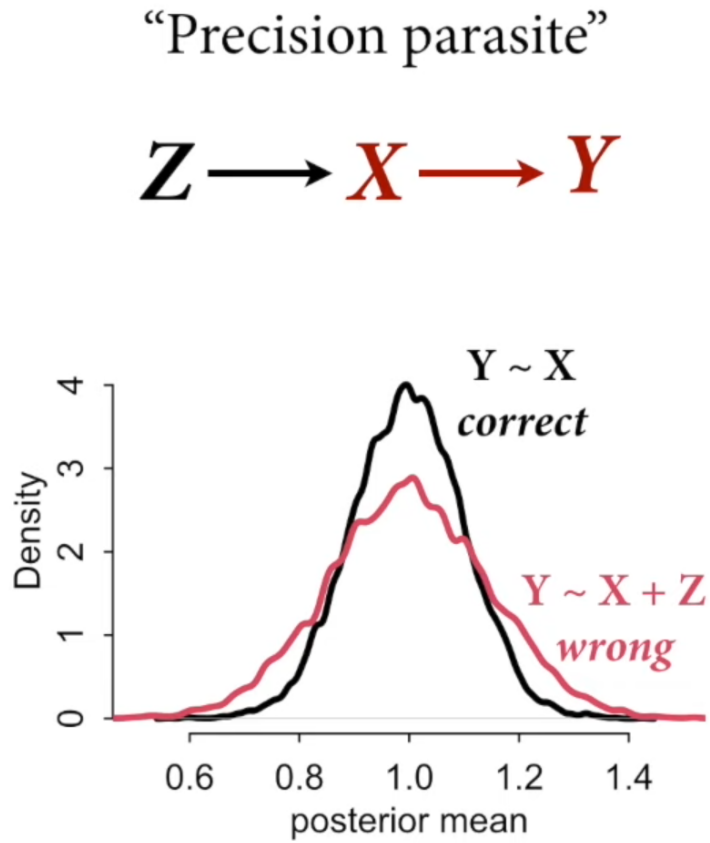
"Precise parasite"

A simulation

```
f <- function(n=100,bZX=1,bXY=1) {
    Z <- rnorm(n)
    X <- rnorm(n, bZX*Z )
    Y <- rnorm(n, bXY*X )
    bX <- coef( lm(Y ~ X) )['X']
    bXZ <- coef( lm(Y ~ X + Z) )['X']
    return( c(bX,bXZ) )
}

sim <- mcreplicate( 1e4 , f(n=50) , mc.cores=8 )

dens( sim[1,] , lwd=3 , xlab="posterior mean" )
dens( sim[2,] , lwd=3 , col=2 , add=TRUE )
```



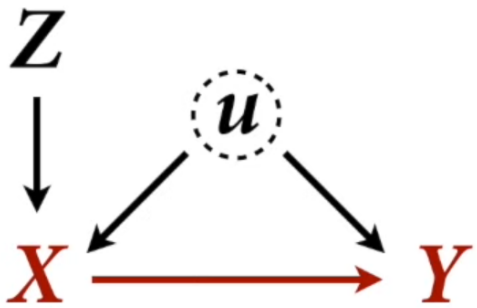
Here you reduce the statistical efficiency. hence the name.

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"Precise parasite"

A more elaborate case

- Now we are going to see bias amplification, because
- X and Y are confounded by u.



- Something awful happens when we add Z.
- We cannot get an unbiased estimate $X \rightarrow Y$ with what we know. Unless we know u.

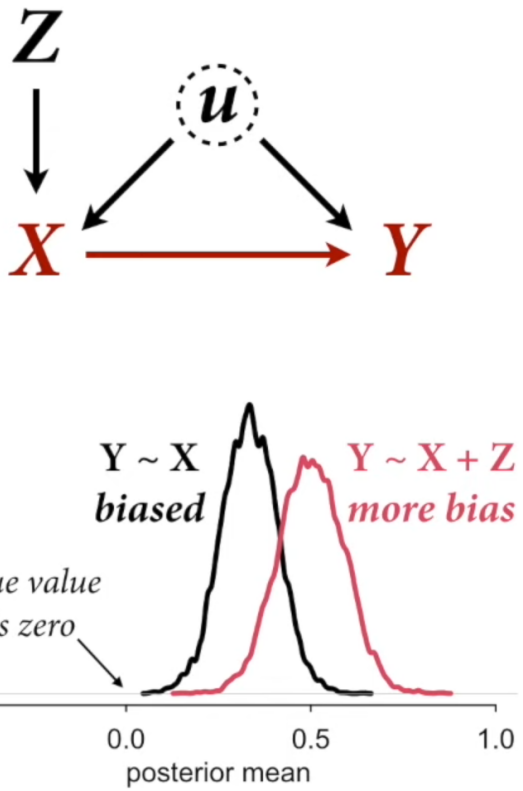
Bias amplifier

A simulation

```
f <- function(n=100,bZX=1,bXY=1) {
    Z <- rnorm(n)
    u <- rnorm(n)
    X <- rnorm(n, bZX*Z + u )
    Y <- rnorm(n, bXY*X + u )
    bX <- coef( lm(Y ~ X) )['X']
    bXZ <- coef( lm(Y ~ X + Z) )['X']
    return( c(bX,bXZ) )
}

sim <- mcreplicate( 1e4 , f(bXY=0) , mc.cores=8 )

dens( sim[1,] , lwd=3 , xlab="posterior mean" )
dens( sim[2,] , lwd=3 , col=2 , add=TRUE )
```



Causes of the exposure are typically bad news in a regression model.

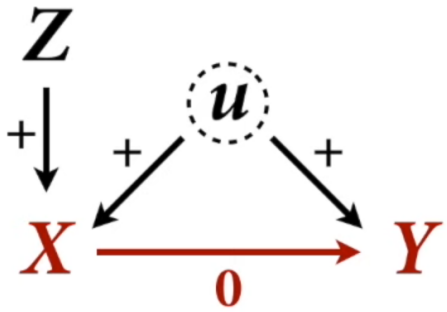
Bias amplifier

Tentative explanation

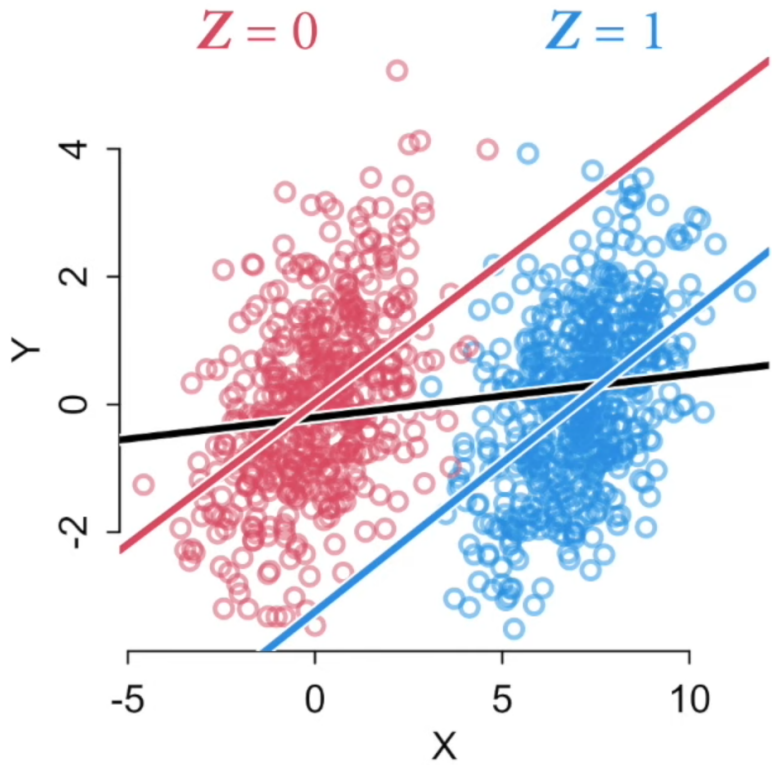
- Covariation in X and Y requires variation in their cause(s); two things cannot **co**vary if they do not **vary** in the first place!
- Within each level of Z (adding a variable to a regression model is stratifying when you look at the effect of another variable), there is less variation in X.
- The confound u is relatively more important within each stratified value of Z.

Bias amplifier

Continuous simulation




```
n <- 1000
Z <- rbern(n)
u <- rnorm(n)
X <- rnorm(n, 7*Z + u )
Y <- rnorm(n, 0*X + u )
```

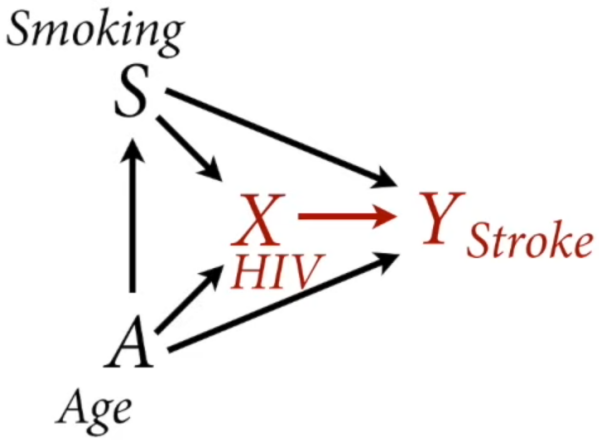


Good and bad controls

- "Control variable": variable introduced to an analysis so that a causal estimate is possible.
- We have seen heuristics fail so far in this lecture: adding a control variables can worsen the estimate compare to omitting it.
- Make assumptions explicit.

Not all coefficients are causal

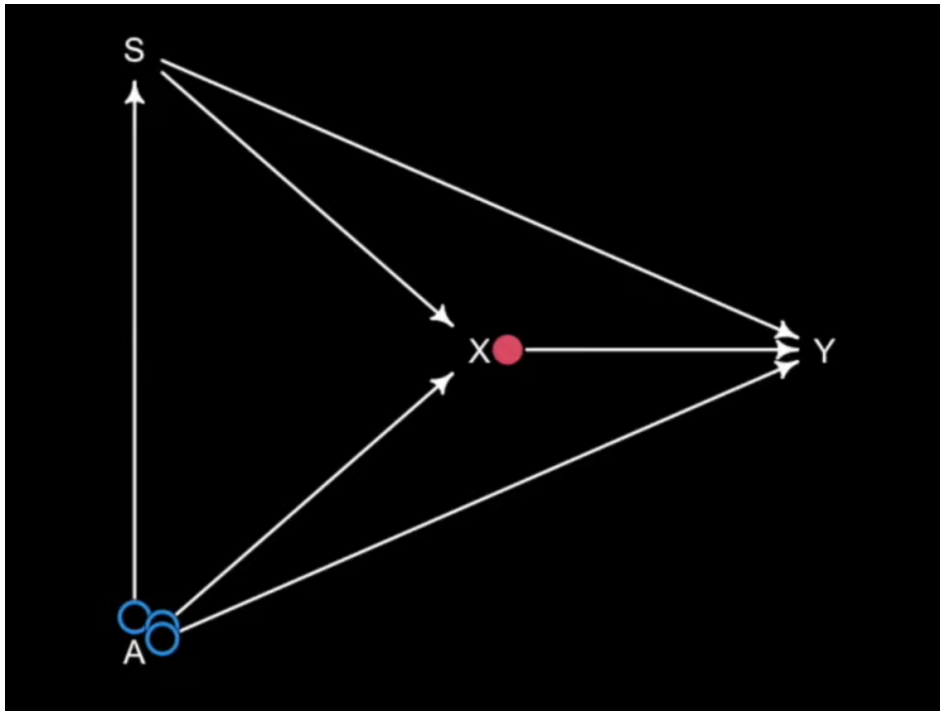
- Not all coefficients are causal effects; statistical models designed to identify $X \rightarrow Y$ will not also identify effects of control variables: even when needed they may only be added to close backdoor path. The coefficients might not be causal.
- An example:



(from Westreich and Greenland 2013, "The Table 2 Fallacy")

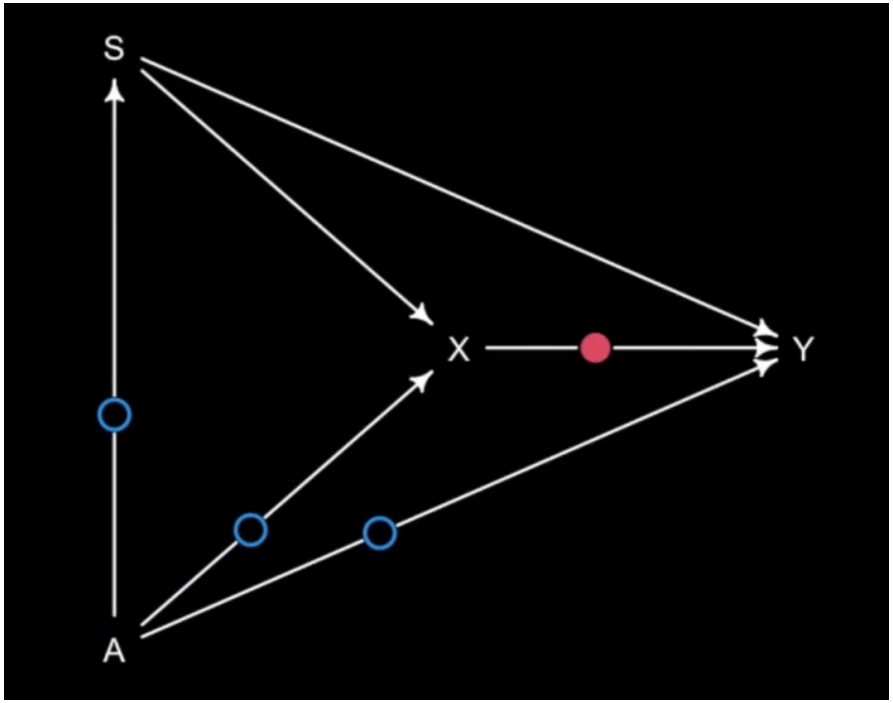
HIV → Stroke example

Animation



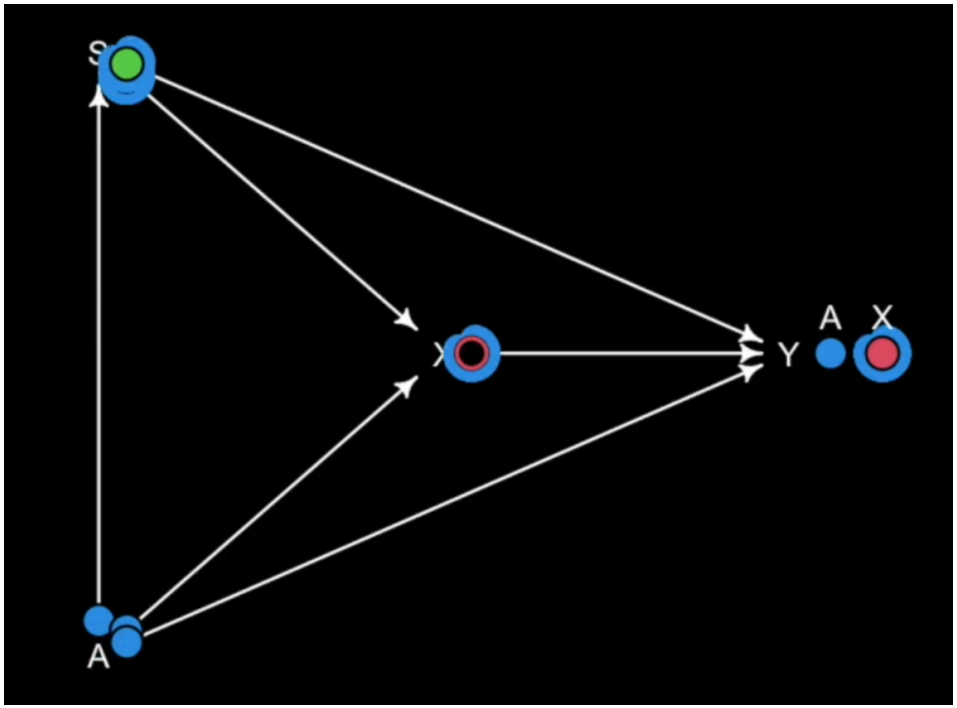
HIV → Stroke example

Animation



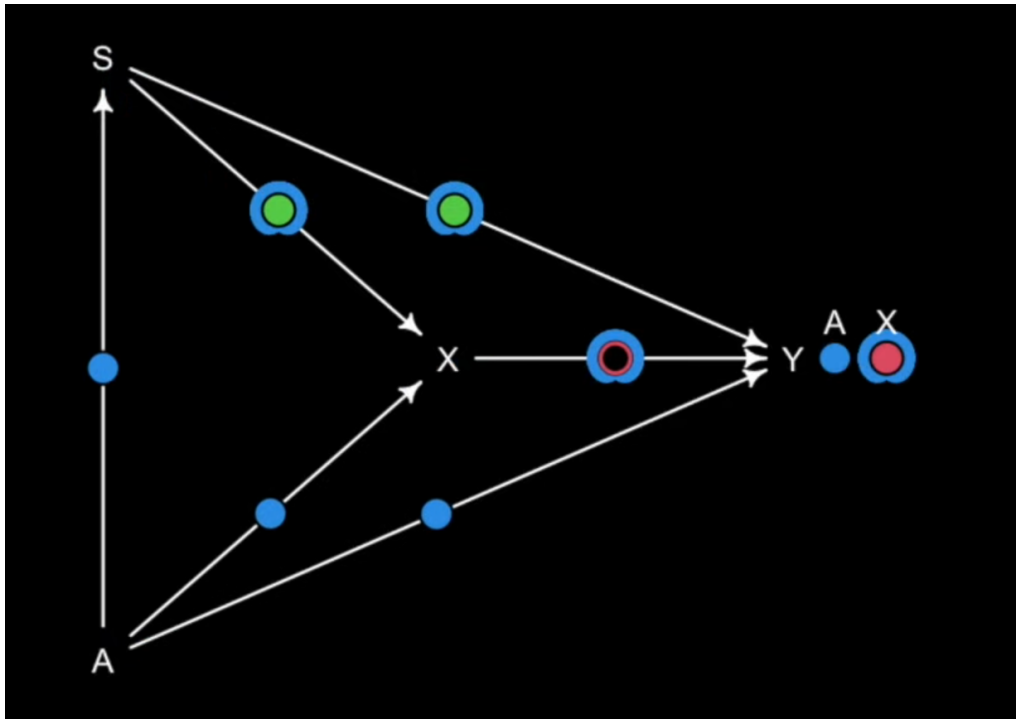
HIV → Stroke example

Animation



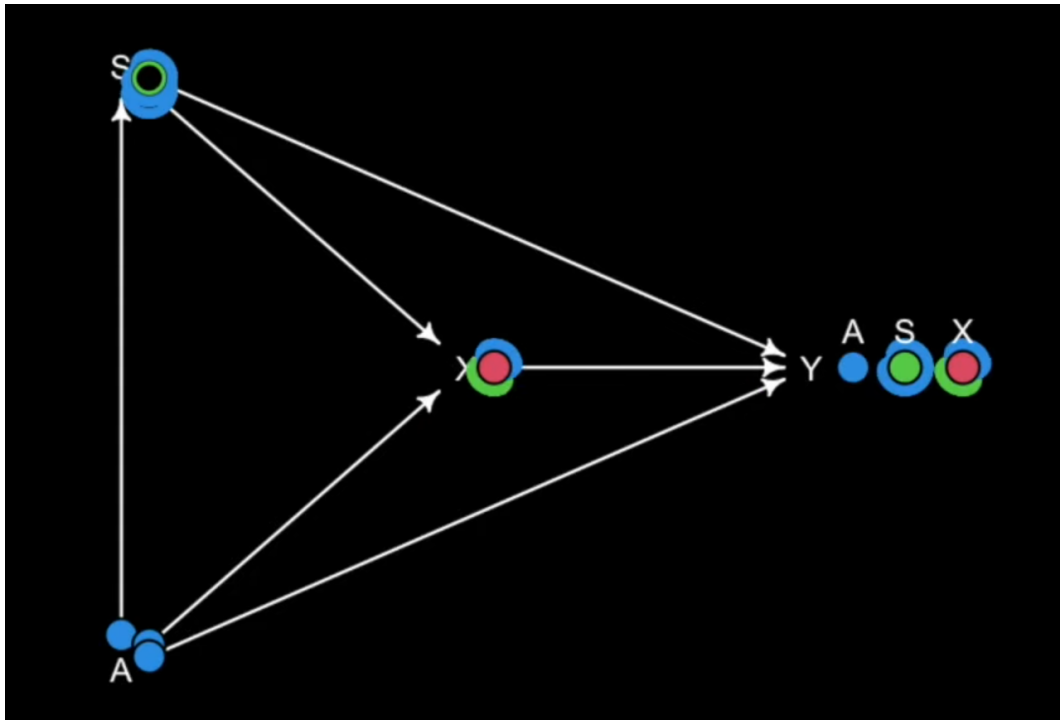
HIV → Stroke example

Animation



HIV → Stroke example

Animation



HIV $\rightarrow$ Stroke example

Use the backdoor criterion

- Goal: "decontaminate" X from the other effects ("crown")
- List all the paths from X to Y: $X \rightarrow Y$, $X \leftarrow S \rightarrow Y$, $X \leftarrow A \rightarrow S \rightarrow Y$, $X \leftarrow A \rightarrow Y$
- To close the (three) backdoor paths, we need to control for A and S. Assuming the DAG is true, we can estimate the $HIV \rightarrow Stroke$ effect:

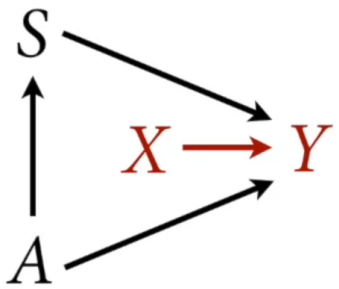
$$
\begin{aligned}
Y_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta_X X_i + \beta_S S_i + \beta_A A_i
\end{aligned}
$$

- The β_X coefficient is by essence causal measuring the $X \rightarrow Y$ direct effect. We simulate a $\text{do}(X)$. Notice we still need to marginalise for a target population, i.e. pick up values for A and S, as they are also risk factors.

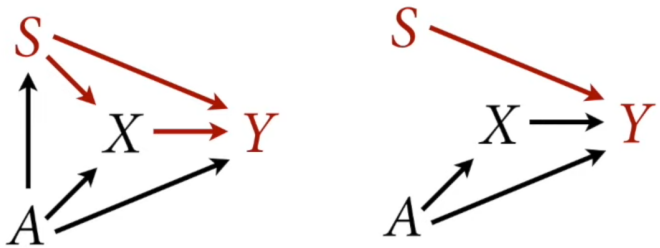
HIV $\rightarrow$ Stroke example

Coefficient analysis

$\mathrm{do}(X) \rightarrow Y$ effect by conditioning on A and S.



- With the same DAG, the unconditional effect $S \rightarrow Y$ is confounded by A (left).

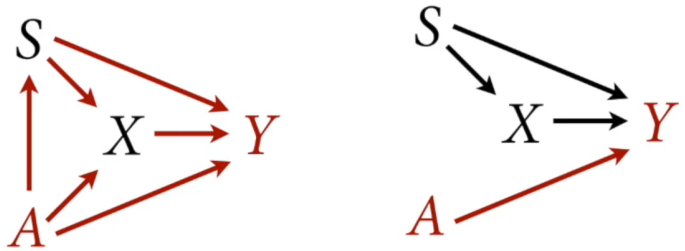


When conditioning on A, we get the direct effect of $S \rightarrow Y$: interesting but not the same meaning as the $X \rightarrow Y$ coefficient.

HIV $\rightarrow$ Stroke example

Coefficient analysis

- The (unconditional) total causal effect $A \rightarrow Y$ flows through all paths (left). We only need to regression Y on A!?!



When conditioning on A and X, we are only left with the direct $A \rightarrow Y$ causal effect (right). Might be tiny.

- Message: if you do ONE table with the regression coefficients, you may mislead your reader! You need to present your causal model along with all the coefficients and their explicit interpretation.

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Conclusive remarks

Imagine confounding

- Often we cannot credibly adjust for all confounding. Don't give up.
- Biased estimates might be better than none. As long as you are aware or suspect them.
- Sensitivity analysis (later in the course): draw the implications of what you don't know.
- Find natural experiments or design one.